

Gabriel Cha

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EDUCATION

Columbia University

M.S. in Data Science, 3.53 GPA

New York, NY

09/2025 – 12/2026

- Applied Large Language Models & Agentic AI, Distributed Compute & LLM Training, Deep Learning & Neural Networks

University of California, San Diego

B.S. in Data Science, 3.75 GPA

San Diego, CA

09/2021 – 05/2025

- Teaching Assistant: Taught data science (DSC 10, DSC 80) for 9 quarters to 1,200+ students, holding 120+ office hours

EXPERIENCE

Honeywell

AI/ML Engineer Intern

Atlanta, GA

05/2026 – 08/2026

- Architected the AI-facing interface for Honeywell's physical-security platform: an MCP (Model Context Protocol) server—13 tools—that lets AI agents operate security systems (credentials, access, lockdowns) through one unified interface
- Engineered a shared ontology that fuses fragmented data from multiple vendor backends into one clean, machine-consumable schema—so agents stay portable when a new system is added, with no rewrites to agent-facing tools
- Benchmarked the ontology against raw API docs in a controlled test: grounding agents in the structured ontology cut token usage 34% and dropped completion of the same task from 7 agent turns to 2 on average
- Developed `llms.txt`, a knowledge harness that helps AI coding agents (Cursor, Claude Code) generate correct API calls without hallucinating parameters or auth flows—reducing token usage 60%

Root Access AI (venture-backed AI startup)

AI/ML Engineer Intern

Manhattan, NY

09/2025 – 05/2026

- Engineered production LangChain coding agents for firmware, grounding agent reasoning in hardware datasheets to generate accurate device-control code; helped grow the enterprise customer base from 3 to 7 on a 5-person team
- Designed a “Plan Mode” planning-then-execution workflow enabling structured reasoning, where the agent autonomously runs grep-based codebase exploration to ground its plan before writing any code — improving alignment with user intent
- Increased the agent's hardware-specific accuracy by 34% by fine-tuning open-weight LLMs via QLoRA on cloud A100s using a custom-curated dataset

DSTL Lab

AI Research Assistant

San Diego, CA

10/2024 – 05/2025

- Designed an AI chatbot that generates course-aligned practice problems by retrieving and processing course-specific materials against active learning objectives; co-authored a SIGCSE 2026 paper on the work
- Raised LLM generation accuracy from 23% to 87% by designing a two-stage data pipeline that cleaned and structured raw educational content into optimized model inputs

Edison

Data Science Intern

Los Angeles, CA

06/2024 – 09/2024

- Built a scalable system monitoring 70+ solar plants for automated outlier detection, transforming a 3-day manual review into a 10-minute automated workflow and preventing \$5K+/month in losses from undetected faults
- Developed a Python ETL pipeline processing 1TB+ across 100+ files, eliminating 4-day manual reporting cycles to accelerate data delivery for energy-trading clients
- Collaborated with a cross-functional energy analytics team to implement detection algorithms, improving system reliability and enabling faster grid-optimization decisions

PROJECTS

Cloi.ai: AI debugging CLI (Y Combinator Interviewed)

05/2025

- Built an end-to-end developer tool integrating LLMs/SLMs for personalized debugging recommendations; packaged to npm and iterated based on user adoption metrics

Classifying Emotions with Machine Learning

09/2023 – 06/2024

- Led \$6,500 grant-funded research classifying emotions in human speech, training and benchmarking multiple machine learning models to identify the top performer

SKILLS

AI & Agents: LangChain, LangGraph, LLMs, RAG, Fine-Tuning (QLoRA), Context Engineering, LLM-as-a-Judge

Machine Learning: PyTorch, TensorFlow, Hugging Face, Scikit-learn, NLP, Distributed Training

Languages & Data: Python, SQL, Java, Git, Pandas, Spark, NumPy, Tableau, ETL, A/B Testing